

1. Problem definition:

1.1. What is the Problem?(note: possible names include - compass

Throughout their studies, students often lack clarity on what to pursue beyond compulsory coursework i.e. which electives, minors, extracurriculars, exchanges, internships, projects, etc. Consequently, many reach the end of their degree in a state of limbo, unable to determine how to progress further in their field, how to pivot to another, etc. often realising late that valuable opportunities had been available to them from the outset of their studies. Such choices directly shape a student's competitiveness for subsequent career paths, including postgraduate study or initial employment. Notably, the elements needed to construct such a path already exist, what is lacking is a means of integrating them into a coherent structure. Hence, relieving students of the burden of undertaking this synthesis unaided. Both hold this problem, at different levels. Students experience it directly; the university owns it institutionally, since it employs the advisers students turn to, carries the cost of study delay, and is assessed externally on graduate employability and satisfaction. The system is commissioned by the institution and used by the students, and Section 3 separates the objectives along that line.

1.2. Subproblems

1.2.1. Outcome visibility gap

Students often see individual opportunities without really understanding what those choices will lead them to, which leads them to making choices without properly evaluating its long term outcome.

1.2.2. Lack of consolidation

Relevant opportunities are spread across university websites, courses, student associations, career portals, university databases, and informal sources.

1.2.3. Delayed feedback

Students discover whether their academic choices were useful much later. At that point it is usually too late and difficult to acquire the correct knowledge and skills.

1.2.4. Interest-to-choice gap

Students can have broad interests that they want to focus on, but they often struggle to put these interests into academic choices.

1.2.5. Changing goals and path adaptability

During studies, students frequently change their ambitions and interests. However, existing planning tools treat study planning as static rather than continuously reassessing what options remain available when a student changes direction.

2. Proposed solution

A study-path advisor sitting between a student's stated interests and the choices their programme still permits. Rather than presenting opportunities as a catalogue to browse, it works backwards from where the student wants to end up: it interprets a loosely described interest or destination, establishes what that destination demands, compares this against what the student has completed and can still fit into their remaining time, eliminates combinations the regulations make impossible, and returns a few coherent whole paths, each stating what it opens and closes off, re-evaluating whenever the student's direction changes. Interpreting destinations and opportunities is the part genuinely requiring intelligence: both are inconsistent free text written for different purposes, and no fixed rule maps one to the other. Feasibility, by contrast, follows institutional rules and requires no learning. The student remains the decision-maker; the system narrows and explains, but does not enrol, apply or decide, and presents no path as a guarantee. A deterministic solution was rejected for the interpretation step: it needs a fixed mapping from interests to opportunities, and none exists or can be maintained by hand. Course descriptions and skill taxonomies share almost no vocabulary — matched against ESCO, exact string matching resolves only 4% of the distinct skill phrases in our course data, and under 8% even when all alternative labels are included, leaving over 92% of distinct phrases unresolved. Feasibility is the opposite case: prerequisites, credit rules and deadlines are fully determined by regulation, so they stay rules rather than being learned.

Input from	What it provides	Output	Content
Student	Stated interest or destination in their own words; completed courses, programme, year, remaining credits; constraints (language, workload, exchange)	Ranked shortlist	Three to five coherent paths, each feasible in the remaining study time

Institution	Descriptions and learning outcomes of courses, minors, projects, internships, exchanges	Outcome statement	Which destinations each path opens — master's programmes, research areas, roles
Institution	Study regulations: prerequisites, credit rules, compulsory components, decision deadlines	Justification	What each element contributes, and why the paths are ordered as they are
Wider environment	Master's entry requirements; competencies named in graduate vacancy and internship listings	Trade-off warning	What each path makes harder or rules out
The system's own use	Which paths students accepted, rejected, or later moved away from	Timing signal	Which decisions must be made in which period, before the option expires

3. Goals/ Objectives:

3.1 Organisational objectives — the benefit to the institution. Reduce the advising effort spent on routine elective and minor questions, shorten the study delay caused by late or poorly-fitted choices, and strengthen graduate progression into master's programmes and relevant employment — outcomes on which the faculty is externally assessed.

3.2 Leading objectives — how we define success of the system. Achieve adoption ahead of each enrolment window, ensure students complete the sessions they start and open the justification behind a recommendation, and bring them back when their direction changes — all observable long before the organisational outcomes can be measured.

3.3 User-oriented objectives — what the system delivers against the problem. Enable a student to describe an interest in their own words and receive three to five feasible paths (1.2.4); see what each opens and closes off (1.2.1); see which decisions expire when (1.2.3); and have the options reassessed on a change of direction (1.2.5).

3.4 Component (model) objectives — how good the system must be. Relate a stated interest to the courses that serve it at an accuracy defensible against a baseline; never propose a path the regulations forbid; explain every recommendation; respond fast enough for interactive use; and process student data lawfully. Section 4 gives the thresholds.

4. Measurements/KPIs

Each KPI is linked to the objective it measures, with its definition and how it is measured. Targets are initial values, to be adjusted once a baseline has been measured.

Objective	Measure / KPI	Target	How it is measured
3.1	Adviser time spent on routine elective and minor questions	– 30%	Adviser time logs, against the semester before deployment
3.1	Students reporting a clear plan for their remaining electives	≥ 70%	End-of-semester survey, single item, all enrolled students
3.2	Students using the system during an enrolment window	≥ 40%	System logs: unique users divided by eligible students
3.2	Recommendations where the justification is opened	≥ 50%	Interaction logs
3.3	Interest fit, rated by the student after a session (1–5)	≥ 4 avg	In-session survey item, averaged across respondents
3.3	Change in decision confidence, before vs after a session (1–5)	+ 1.0	Paired survey item asked at the start and the end of the same session
3.4	Recommended electives/minors matching a manually checked sample	≥ 0.85	Held-out labelled sample, compared against a string-matching baseline
3.4	Correctly tagged courses within the top 5 recommendations	≥ 3 of 5	Manual check of the top 5 against the student's stated interest